

Hospital Readmission Prediction (Machine Learning)

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Introduction:

Hospital readmissions within 30 days are a major operational and financial challenge for healthcare systems.

Predicting which patients are at high risk enables:

- Better resource planning
- Early interventions
- Reduced pressure on emergency services
- Improved patient outcomes

This phase focuses on building a machine learning model to predict 30-day readmission using the UCI Diabetic Readmission dataset.

Dataset link:

UCI Diabetic Readmission Dataset

<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

Dataset Overview

The dataset contains 101,766 hospital encounters for diabetic patients across 10 years.

Key Characteristics:

- Rows: 101,766
- Columns: 50
- Target variable: readmitted_30
- Positive class: ~11% (highly imbalanced)
- Feature types:
 - Demographics
 - Diagnosis codes (ICD-9)
 - Medication changes
 - Visit history
 - Lab results

Data Protection & Privacy:

This project uses publicly available, fully anonymised datasets. No personal or identifiable patient information was accessed, so data protection regulations such as GDPR and the UK Data Protection Act do not apply.

Challenges

- Missing values represented as "?"
- High-cardinality categorical variables
- Imbalanced target
- No clinical notes or time-series data

Exploratory Data Analysis (EDA)

Key Findings

- 37 categorical variables
- Several columns with >80% missing values
- Diagnosis codes contain hundreds of unique ICD-9 categories
- Target imbalance:
 - 0: 90,409
 - 1: 11,357

Pre-processing & Feature Engineering

Handling Missing Values

- Replaced "?" with NaN
- Dropped columns with extremely high missingness
- Filled categorical missing values with "Unknown"

Target Engineering

Converted original readmitted column into binary:

<30 → 1

NO → 0

>30 → 0

Encoding

- Converted categorical columns to category dtype
- Applied one-hot encoding
- Cleaned column names for XGBoost compatibility

Scaling

- Applied StandardScaler to numeric features
- Ensured target column remained untouched

Saving the Processed Dataset

The final ML-ready dataset was exported for model training.

Saved to:

data/processed/processed_diabetic_data.csv

```
> \n\n    print("Processed dataset saved to: data/processed/processed_diabetic_data.csv")\n\n[20]  ✓  0.0s  Python\n\n...  Processed dataset saved to: data/processed/processed_diabetic_data.csv
```

Fig 1. Processed dataset saved to the project directory

Model Training

Models Evaluated

- Logistic Regression
- Random Forest
- XGBoost (Best Performer)

Best Model Performance (XGBoost)

Metric	Score
Accuracy	0.889
ROC-AUC	0.693
Precision	0.633
Recall	0.016
F1 Score	0.032

Interpretation

- High accuracy due to class imbalance
- Low recall indicates difficulty identifying positive cases
- ROC-AUC of 0.693 shows moderate predictive power

Saved Model

models/xgb_readmission.pkl

```
import joblib\njoblib.dump(xgb, "../models/xgb_readmission.pkl")\n\n[21]  ✓  0.0s  Python\n\n...  ['../models/xgb_readmission.pkl']
```

Fig 2. Final trained model saved as a .pkl file for deployment and evaluation.

Model Evaluation

Confusion Matrix

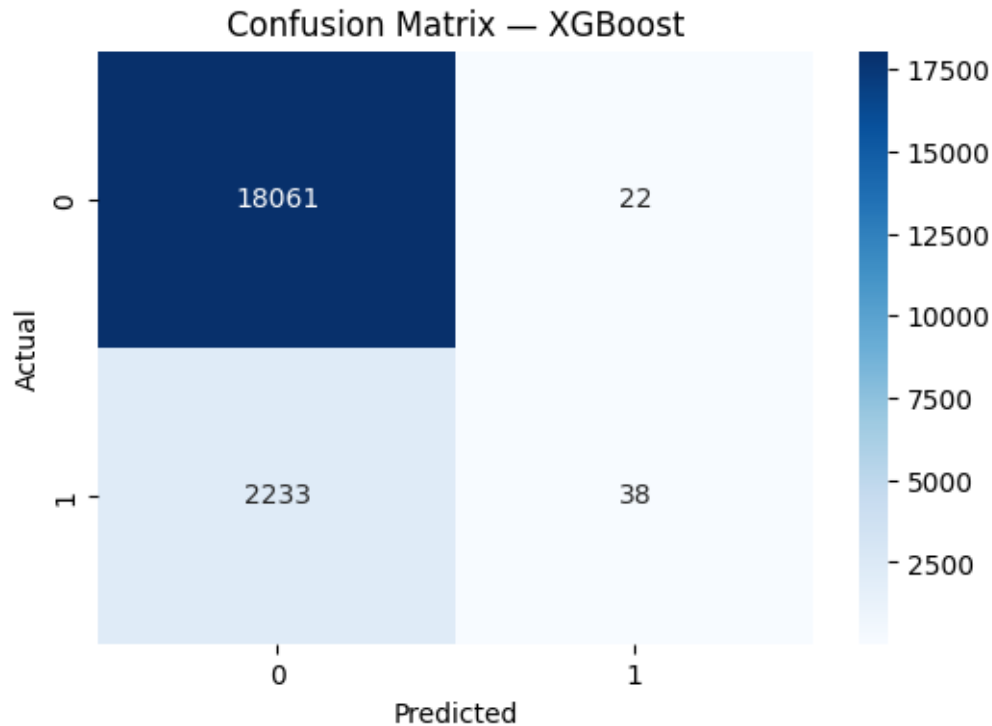


Fig 3. Confusion Matrix showing model performance on test data

Shows strong prediction of majority class but limited detection of readmissions.

Classification Report

- Precision is acceptable
- Recall is low
- F1 score reflects imbalance

	precision	recall	f1-score	support
0	0.89	1.00	0.94	18083
1	0.63	0.02	0.03	2271
accuracy			0.89	20354
macro avg	0.76	0.51	0.49	20354
weighted avg	0.86	0.89	0.84	20354

Fig 4. Classification report with precision, recall, and F1-score

ROC Curve

XGBoost achieved the highest AUC (~0.693).

```
from sklearn.metrics import roc_auc_score

auc_score = roc_auc_score(y_test, y_prob)
auc_score
```

✓ 0.0s Python

0.6925412647240172

Fig 5. ROC curve showing model discrimination ability.

Precision-Recall Curve

Confirms difficulty in identifying minority class.

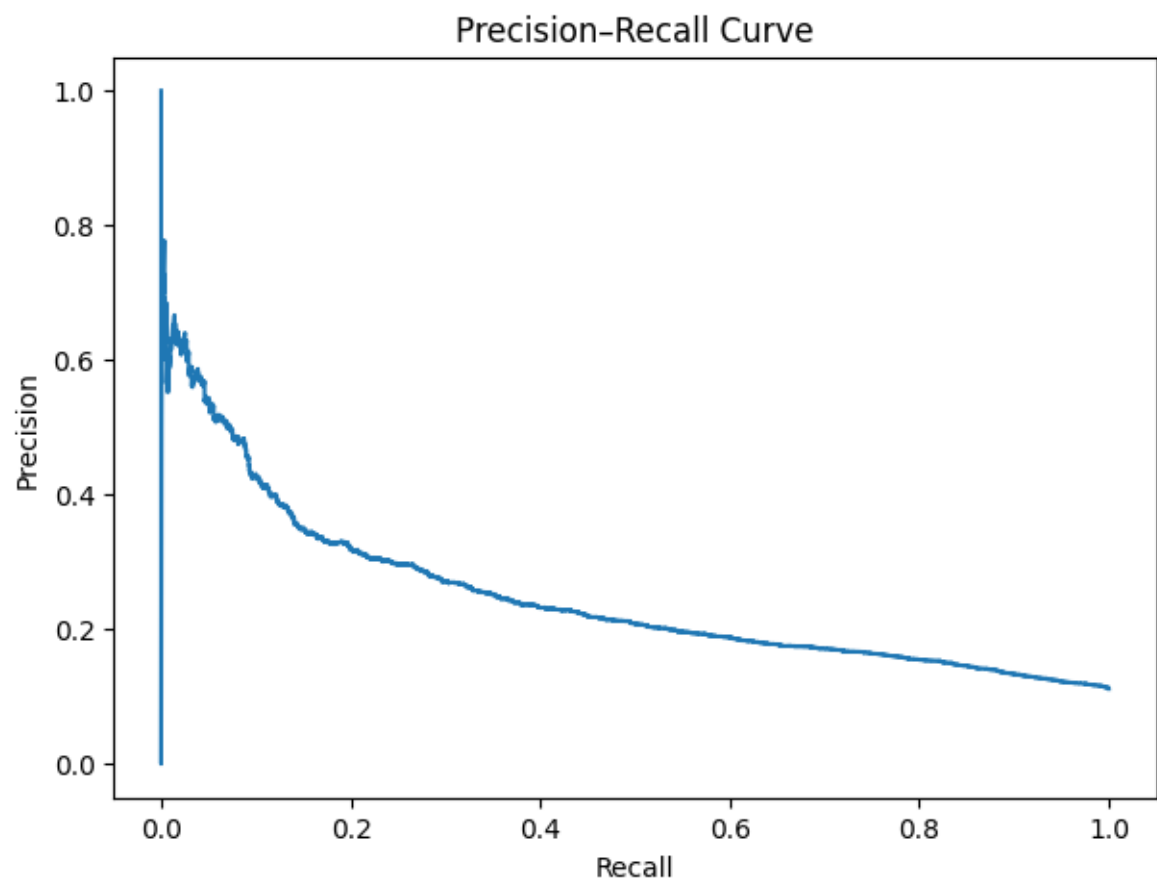


Fig 6. Precision-Recall curve for imbalanced classification.

Calibration Curve

Model moderately calibrated but under-predicts risk.

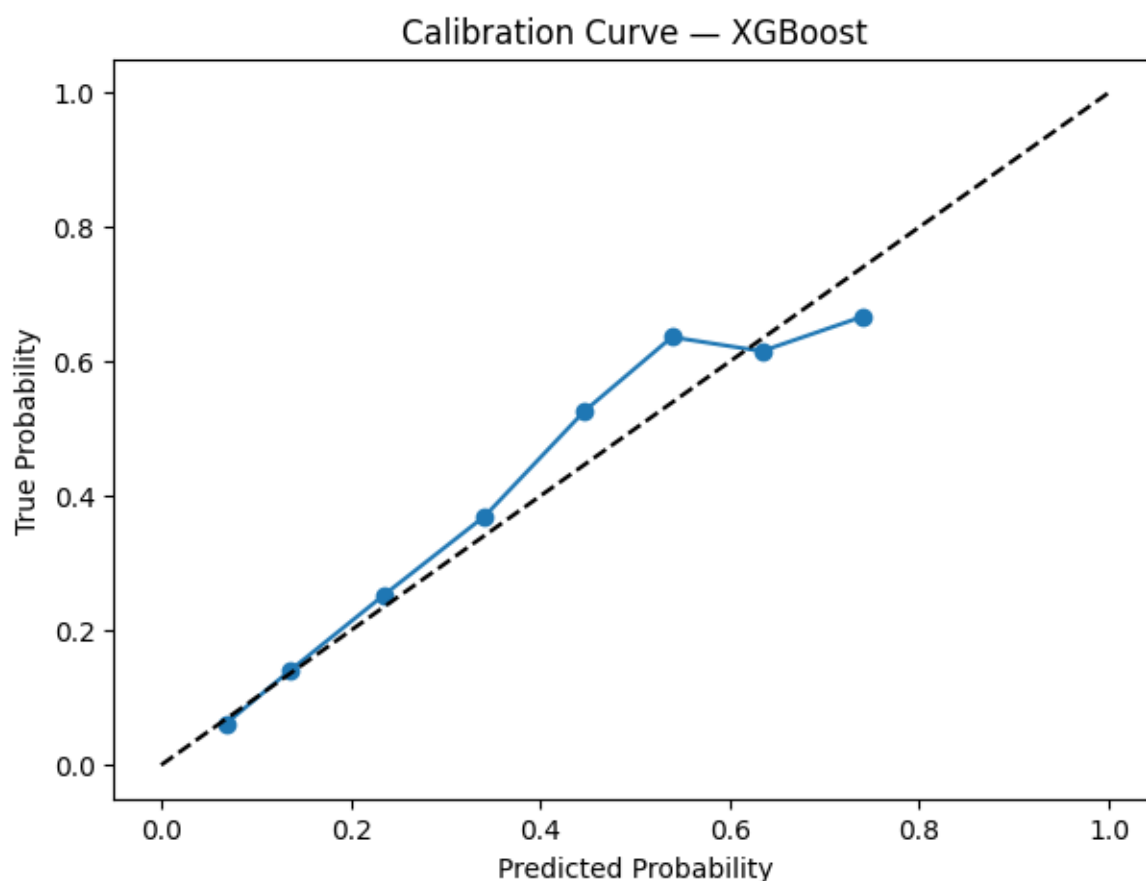


Fig 7. Calibration curve showing probability calibration.

Feature Importance

Feature importance was not included because the evaluation notebook focused on performance metrics and probability-based evaluation plots.

Limitations

- Severe class imbalance
- High-cardinality diagnosis codes
- Dataset originates from US hospitals (not NHS)
- No clinical notes or time-series data
- Limited recall for positive class

Future Improvements

- Apply SMOTE or class-weighting
- Hyperparameter tuning for XGBoost
- Add SHAP explainability
- Deploy via FastAPI
- Integrate predictions into Power BI
- Explore NHS-specific datasets

Conclusion

Phase 2 successfully delivers a complete machine learning pipeline for predicting 30-day hospital readmission.

Combined with Phase 1 (NHS A&E Dashboard), this project demonstrates:

- End-to-end data engineering
- BI reporting
- Predictive modelling
- Healthcare analytics capability